

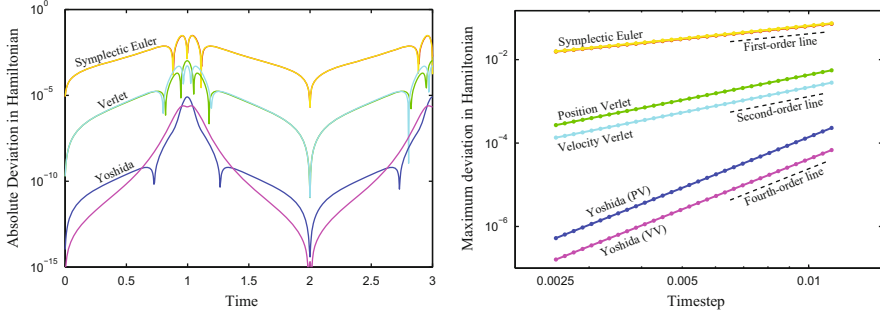


Fig. 3.2 We compare the symplectic Euler, Verlet and Yoshida schemes in application to a Lennard-Jones oscillator. The plot shows the absolute deviation in the computed Hamiltonian (*left*) as a function of time, for each scheme at a fixed timestep $h = 0.005$. Moreover we simulate the system using different stepsizes and compute for each stepsize the maximum deviation in the Hamiltonian (*right*), comparing the results with guide lines associated to various powers of the step size. The scheme used as the “base” method for the Yoshida composition methods is denoted in the parenthesis (either position or velocity Verlet)

as their base second-order methods (the base method is denoted in parenthesis in Fig. 3.2), giving similar results.

Of course we need not stop at a fourth-order scheme: we can use the constructed method to target any higher order terms we wish by composing the new scheme in a symmetric form and proceeding identically to the above.

These methods are not the best possible choices among schemes that alternate solving the terms of the splitting [260, 277], either in the sense of having the greatest stability or in the sense of having the smallest leading error coefficients. We can view the Suzuki-Yoshida methods as one type of general composition scheme [260, 277, 326]:

$$\begin{aligned} \mathcal{F}_h = & \exp(\alpha_1 h \mathcal{L}_T) \circ \exp(\beta_1 h \mathcal{L}_U) \circ \exp(\alpha_2 h \mathcal{L}_T) \circ \exp(\beta_2 h \mathcal{L}_U) \circ \cdots \circ \\ & \exp(\alpha_k h \mathcal{L}_T) \circ \exp(\beta_k h \mathcal{L}_U). \end{aligned}$$

The coefficients of these methods can be optimized according to many criteria, for example to reduce the coefficients associated to certain types of elementary differentials appearing in the leading error term [260], or to have maximal performance for particular families of applications, such as celestial mechanics [37, 127]. For many applications, for example celestial mechanics, such high order methods, and improved variants, are extremely useful, but in the setting of molecular dynamics, it is often found that second-order methods provide a ‘good enough’ option for numerical integration, the computational price-tag of higher order methods outweighing potential numerical gains. After all, the potential energy functions used in the ODEs are not perfect reflections of the laws of physics, and past a certain point using a high order method merely finds a better solution to an approximate model. Furthermore, most molecular simulations are performed at large timestep,